

Brian Tsoi

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Education

University of Toronto

Bachelor of Applied Science in Computer Engineering

Expected Oct 2026

GPA: 3.65/4.00

Relevant Coursework: Operating Systems, Algorithms & Data Structures, Computer Networks, Distributed Systems, Introduction to Artificial Intelligence, Applied Fundamentals of Deep Learning

Experience

Tenstorrent

AI Compiler Engineer Intern - Pytorch/XLA

Jan 2026 – Present

Toronto, Canada

- Built **tensor parallel** inference benchmarking framework in **Python**, enabling continuous performance tracking across **17+ production models** (Llama, Qwen, Mistral, Falcon) in CI
- Implemented **end-to-end** TP generative inference for **GPT-OSS-20B** on 8/32 chips systems
- Developed **MLIR** compiler pass to convert unsupported operations in **XLA** graphs into target IR

NVIDIA

System Software Engineer Intern - NVStreams (DriveOS)

May 2025 – Aug 2025

Santa Clara, CA

- Engineered a multi-threaded **C++** stress testing framework to validate NVStreams data pipeline, emulating behavior of **30+** high-throughput data streams across **90+ parallel processes**
- Resolved **stability regression** in the QNX automotive RTOS by leveraging custom stress test
- Reduced a **4-hour** manual performance data collection cycle to **20 minutes** using **Python** and **Bash**

Mozilla

Backend Software Engineer Intern - OS Integration for Firefox

Sep 2024 – Apr 2025

Toronto, Canada

- Authored and open-sourced memtest, a **low-level Rust** library with **15+ memory testing algorithms**, achieving **over 4,800 downloads** in its first 3 months on crates.io
- Invalidated **1.16% main process crashes**, preventing wasted engineering hours on non-reproducible failures, by integrating memtest into Firefox's crash reporter to identify faulty hardware
- Analyzed **over 108k** crash reports with **Python** and **SQL** for memory testing effectiveness evaluation

Mozilla

Backend Software Engineer Intern - OS Integration for Firefox

May 2024 – Aug 2024

Toronto, Canada

- Reduced volume of targeted crash types by **5%** in a **2.3M+/month** crash pipeline, by implementing **x86-64 ASM semantic validation** in **Rust-minidump** analyzer to identify invalid reports
- Automated early testing on over **25k reports** with **Python** to ensure validation feature reliability

Projects & Extracurriculars

ESP32 Microcontroller-based Arcade Machine

Sep 2024 – Apr 2025

- Spearheaded a **team of 10** as Project Lead in end-to-end **C++** and **FreeRTOS** development of a dual-ESP32 game system, integrating **11 I/O peripherals** and **96 LEDs** on a 32"x28"x68" machine

AI Ray Tracing Image Denoising Filter

May 2023 – Aug 2023

- Trained a **PyTorch** autoencoder neural network for denoising low-sampling rate ray tracing images
- Optimized to reach **83% image quality improvement** and outperform conventional filters by **47%**

Skills

Programming: C, C++, Python, Rust, Java, Go, x86-64 Assembly, HTML, CSS, JavaScript, SQL, Git
Embedded/Hardware: STM32 microcontrollers, RTOS, Arduino, Raspberry Pi, UART, I2C, SPI